BUSINESS INTELLIGENCE · SUPPLY CHAIN

Supply Chain Performance & Decision Intelligence

Action Priority Index for customer prioritization, KPI interpretation and operational investigation.

DBT · POSTGRESQL · TABLEAU · PYTHON

THOMAS RABE
Neue Fische, Data Analytics & AI Bootcamp, Capstone Project 2026



One Supply Chain, Three Decision Questions

THE SUPPLY CHAIN FLOW

Customer Order **9,215** → Plant **19** → Origin Port → Freight Lane **22** plant-port links → Delivery

3 KPIs, 3 points of the chain

Capacity Risk | Freight Cost Exposure | Delivery Reliability

STRATEGIC VIEW

Critical Dependencies

- Focus on plants & ports
- 50% capacity weight

DECISION QUESTION

Where are the most critical dependencies?

TACTICAL VIEW

Customer Priority Drivers

- 40% freight
- 20% capacity
- 40% delivery

DECISION QUESTION

How do priorities balance?

OPERATIONAL VIEW

Actions to Review

- Product level focus
- Alternative plant
- 60% delivery weight

DECISION QUESTION

Which actions to review?

Key context: single snapshot, 2013-05-26 — the dashboard reads one day of this chain, it does not simulate or forecast it.

Dataset & EDA: Four Findings Shaped the Design

DATASET

7 / 9,215 / 46 / 19

tables / orders / customers / plants

- Cleaning: 3 duplicate rates + 2 zero-weight orders
- No missing values

Source: Dzalbs & Kalganova (2019), Brunel University London / Figshare.

FINDING 1

1 date

(2013-05-26)

- Only 1 unique date
- Snapshot model (no trend/forecast)

FINDING 2

24.1%

of orders (2,218)

- No freight rate
- Exclude from Freight KPI (rate = relative proxy)

FINDING 3 & 4

92.7–100%

plants/ports (single value)

- Top customer ≤ 10.6%
- Customer = right grain (no plant-level ranking)

123 (1.3%)

orders flagged early & late

- Exclude (not netted to zero)
- Smaller sample for Delivery KPI

Source: Dzalbs & Kalganova (2019), Brunel University London / Figshare

Two Data Hypotheses, Tested Before Modelling

H HYPOTHESIS 1

Capacity risk is structurally concentrated in one plant.

- Capacity
- Delivery

EVIDENCE 1

ρ=0.14 · p=0.40

Capacity util. vs delivery lateness (n=44)
not significant

26 / 46

customers share the maximum capacity value

≈93%

of order volume on **PLANT03**

H HYPOTHESIS 2

C46's freight signal comes from a small lane subset, not the whole customer.

- Freight

EVIDENCE 2

0.563 / 0.090

mean freight rate
C46 vs all others

0.048 = 0.048

median rate
the typical order is normal

n=761/6,234

Welch's t-test
order level

LIMITS

H1: limited differentiation, not a plant-driven cause

H2: business cause unconfirmed; skewed rates make the test indicative only

Decision — C46:
retain + flag, not delete. 3 checks found no technical error: not a thin sample, no fan-out, nesting hypothesis rejected (0 rows).

NEXT → *How do these findings become one comparable score?*

dbt Builds One Foundation, Three Weightings Read It

DBT PIPELINE

SRC	STG	PREP	MART
7 raw tables	**7** rename & cast	**4** models: freight · capacity · delivery · kpi_base	**3** models: fpindex · plants · reallocation

DATA QUALITY

- All uniqueness tests passing
- All not_null tests passing
- KPIs stored normalized (0-1)
- Weighting happens in Tableau (stays adjustable)

APINDEX FORMULA

$$\frac{(F \cdot wF + C \cdot wc + D \cdot wD)}{(wF + wC + wD)}$$

THRESHOLDS

Low	Medium	High
<0.30	0.30–0.60	≥0.60

THREE WEIGHTING PROFILES

	EXECUTIVE	SUPPLY CHAIN MANAGER	PLANNER
Freight	20%	40%	20%
Capacity	50%	20%	20%
Delivery	30%	40%	60%

NEXT → *Can the score be trusted — and where does it break?*

Validation Tests the Model: Ranking Holds, Classes Move

SCORING COVERAGE

40/46

customers scored

- 6 not scored due to missing KPI

MODEL AGREEMENT

≈ 0 *(< 0.000000001)*

max deviation (Tableau vs. Python)

- across 40 customers × 3 profiles

DATA INTEGRITY

0/256

duplicates or join fan-out

- 256 reallocation candidates
- across 23 customers

WEIGHT SENSITIVITY

Capacity +10pp

shifts class for 21 / 27 / 24 customers

Strategic	Tactical	Operational
21 / 40	27 / 40	24 / 40

- Smaller shifts for Delivery and Freight

RANKING ROBUSTNESS

Spearman ρ ≥ 0.97

in all 18 scenarios

- ordering is robust
- thresholds are directional

LIMIT

Single snapshot

40 of 46 customers

- robustness is not correctness

NEXT → What does the model show for real customers?

Live Dashboard: One Model, Three Questions

Decision Support: Where to Look First

THE MODEL SUPPORTS



Turning a static snapshot into a ranked, transparent review order.



KEY LIMITS

- 1 snapshot day, no trend
- 24.1% of orders without freight rate
- 6 of 46 customers Not Scored
- 27 of 40 change class at Capacity +10pp
- weights set by judgment, not estimated



STRATEGIC

Which network dependencies to review first



TACTICAL

Which priority drivers to validate before acting



OPERATIONAL

Which alternatives to check next



NEXT STEP

Validate C46 with a domain expert

Questions?

THANK YOU

Thank you for your time

- Freight
- Capacity
- Delivery

NEXT STEPS

Open for questions and discussion

- Feedback
- Ideas
- Your perspective

Explore the interactive dashboard
Scan to view on Tableau Public

Thomas Rabe

Neue Fische
Data Analytics & AI Bootcamp
Capstone 2026